DO NOW — RECALL

Entity relationship diagrams

Name: _____ Class: _____ Date: _____

1. Where is a FK stored in a one-to-many relationship?

2. What does PK stand for?

3. What is an associative entity?

4. A BST root20 has left10 and right30; insert 25. Describe comparisons and placement.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

On the many side.

2. Model answer

Primary key.

3. Model answer

Bridge/junction entity resolving many-to-many.

4. Model answer

25>20 right to30; 25<30 left child of30.

Vocabulary — know these terms

ERD — Entity relationship diagram.

cardinality — Number of related instances.

junction table — Table linking M:N records.

one-to-many — One entity instance can have several related rows.

NEW LEARNING

Read, annotate and explain

Understand

ERDs model entities and relationship cardinalities before implementing tables.

Apply

One-to-many relationships are typically represented with a foreign key on the many side.

Evaluate

Many-to-many relationships require an associative/junction table when translated into a relational schema, often with two foreign keys.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Model Author and Book when one author writes multiple books, each book has one primary author.

Predict first

Each Book row stores one AuthorID

Model answer

several books can refer to the same author.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State meanings of 1:1,1:M and M:N cardinalities. [3]

Question 2

Draw/describe an ERD for Customer and Order with one customer placing many orders. Give foreign key. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Resolve Student many-to-many Course into relational tables; describe keys and integrity. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A many-to-many relationship is usually represented in relational tables by placing one simple FK on either table.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.